SAYILAR

2.1 $\dfrac{1}{2+\dfrac{1}{1+\dfrac{1}{2+\frac{1}{3}}}} = \dfrac{1}{2+\dfrac{1}{1+\frac{3}{7}}} = \dfrac{1}{2+\frac{7}{10}} = \dfrac{1}{\frac{27}{10}} = \dfrac{10}{27}$

$2+\frac{1}{3} \to \frac{7}{3}$; $1+\frac{3}{7} \to \frac{10}{7}$

2.2 $\frac{5}{12} < \frac{x+1}{60} < \frac{1}{2} \to \frac{25}{60} < \frac{x+1}{60} < \frac{30}{60} \to 25 < x+1 < 30$

$24 < x < 29$

$28-25=3$

2.3 $\dfrac{(12:4+2)\cdot(14-6:3)}{2-\frac{2}{3}} = \dfrac{5\cdot 12}{\frac{4}{3}} = 5\cdot 12\cdot\frac{3}{4} = 45$

$2-\frac{2}{3} \to \frac{4}{3}$

2.4 $|1-\sqrt{3}| \to \sqrt{3} > 1$ mutlak değerin içi negatif olamaz

$-|1-\sqrt{3}| = -(1-\sqrt{3}) = \sqrt{3}-1$

2.5 $2 < x < 6$ ise $|2-x|-x+a=11$ $a=?$

$x-2-x+a=11$ $a-2=11$ $a=13$

2.6 $\dfrac{5,6\cdot 10^8 - 14\cdot 10^7}{6\cdot 10^5} = \dfrac{56\cdot 10^7 - 14\cdot 10^7}{6\cdot 10^5} = \dfrac{10^7(56-14)}{10^5\cdot 6} = \dfrac{10^7(42)}{10^5\cdot 6} = 7\cdot 10^2 = 700$

2.7 $(111,01)_2 = 2^2\cdot 1 + 2^1\cdot 1 + 2^0\cdot 1 + \frac{1}{4}\cdot 1 = 7 + \frac{1}{4} = \frac{29}{4}$

2.8 $(241)_5 + (110)_4 = (x)_8$

$[25+5\cdot 4+1] + [16\cdot 1+4\cdot 1] = 71+20 = 91$

$91 \mid 8$; $88 \mid 11 \mid 8$; $3 \quad 8 \mid 1$; 3 ; $x=133$

EBOB - EKOK PDF

1) $(-18)+7-(-6)-11 = -29+13 = -16$

2) $\underbrace{[-24:6]}_{-4}\cdot(-3)+\underbrace{15:(-5)}_{-3} = 12-3 = 9$

3) $(-3)^2 = 9 \quad -3^2 = -9 \quad (-2)^3 = -8$

4) $4-3\cdot\underbrace{[2-(1-5)\cdot 2]}_{10}+\underbrace{18:(-3)}_{-6} = 4-30-6 = -32$

5) $\underbrace{(-3)^2}_{9}-[\underbrace{(-24):3}_{-8}+\underbrace{2\cdot(-5)}_{-10}-\underbrace{6\cdot(2-5)}_{18}]\cdot 2 = 9$

6) $48:6\cdot 2-7+3 = 8\cdot 2-7+3 = 9+3 = 12$

7) $x-\{1-[\underbrace{2x-3(x+1)}_{2x-3x-3}]\}+4(2-2x) = x-\{1+x+3\}+8-8x$

$= x-1-x-3+8-8x = 4-8x$

8) $27\cdot 16-27\cdot 6 = 27(16-6) = 27\cdot 10 = 270$

9) a = tek, b = çift $\Rightarrow$ $a^2+3b-ab$ (tek, çift, çift) — Tek + Çift − Çift = Tek

10) $7+10+13+\ldots+40 \quad \frac{40-7}{3}+1 = 12 \quad \frac{47}{2}\cdot 12 = 282$

11) $x+x+2+x+4 = 66 \quad 3x = 60 \quad x = 20 \quad 20, 22, 24$

12) $10+10 = 20$ — Globally yours $10\cdot 10 = 100$

13) $(243)_5 = (?)_{10} \quad (73)_{10}$

$25\cdot 2+5\cdot 4+3 = 73$

14) $(83)_{10} = (?)_5$

$83 \mid 5$, $-80 \mid 16 \mid 5$, $3 \quad -5 \mid 3$, 1 $\quad (313)$

$25\cdot 3+5$

15) $\frac{6!+5!}{4!} = \frac{6\cdot 5\cdot 4}{4}$

16) $-\frac{5}{6}$ (21), $-\frac{3}{4}$ (13), $\frac{1}{2}$ (6)

17) $\frac{5}{6}+\frac{4}{5}-\frac{2}{3}$ (5) (6) (10)

18) $\frac{7}{4}\cdot\left(\frac{-23}{6}\right)\cdot$

19) $2\frac{1}{5} : 1\frac{1}{3}$

20) $\left(1+\frac{1}{3}\right)-\left(\frac{1}{2}-\right.$

21) $\left\{\left[\left(\frac{1}{2}+\frac{1}{3}\right):\right.\right.$

22) $\frac{1}{1+\frac{1}{1+}}$

23) 2,75 + 0,4

$\frac{275}{100}+\frac{1200}{1000}-$

14) $(83)_{10} = (?)_5$

$63 \mid 5$ $\quad 13 \mid 5$ $\quad 3$

-60 $\quad 10 \mid 5$ $\quad 2$

3 $\quad -5$ $\quad 1$

$(313)_5$

$25 \cdot 3 + 5 + 3 = 83$

15) $(132)_b = 56$

$b^2 + 3b + 2 = 56$

$b^2 + 3b - 54$

$b \quad 9 \quad b = 9$

$b \quad -6 \quad b = 6$

16) $\frac{6! + 5!}{4!} = \frac{6 \cdot 5 \cdot 4! + 5 \cdot 4!}{4!} = \frac{4!(30+5)}{4!} = 35$

17) $\frac{-5}{6}_{(-5)}, \frac{-3}{4}_{(-3)}, \frac{1}{2}_{(6)} \rightarrow \frac{-10}{12}, \frac{-9}{12}, \frac{6}{12}$ $\quad \frac{1}{2} > \frac{-3}{4} > \frac{-5}{6}$

18) $\frac{5}{6}_{(5)} + \frac{4}{5}_{(6)} - \frac{2}{3}_{(10)} = \frac{25 + 24 - 20}{30} = \frac{29}{30}$

19) $\frac{7}{4} \cdot \left(\frac{-23}{6}\right) \cdot \frac{24}{11} = -\frac{161}{11}$

20) $2\frac{1}{5} : 1\frac{1}{3} = \frac{11}{5} : \frac{4}{3} = \frac{11}{5} \cdot \frac{3}{4} = \frac{33}{20}$

21) $\left(1 + \frac{1}{3}\right) - \left(\frac{1}{2} - \frac{3}{4}\right) = \frac{4}{3} - \left(\frac{2}{4} - \frac{3}{4}\right) = \frac{4}{3}_{(4)} + \frac{1}{4}_{(3)} = \frac{16}{12} + \frac{3}{12} = \frac{19}{12}$

22) $\left[\left(\frac{1}{2} + \frac{1}{3}\right) : \frac{5}{6}\right] - \left(\frac{3}{4}\right)^{-1} = \frac{5}{6} \cdot \frac{6}{5} - \frac{4}{3} = 1 - \frac{4}{3} = \frac{-1}{3}$

23) $1 + \cfrac{1}{1 + \cfrac{1}{1 + \frac{1}{2}}} = 1 + \cfrac{1}{1 + \frac{2}{3}} = 1 + \frac{3}{5} = \frac{8}{5}$

24) $2{,}75 + 0{,}48 \cdot 2{,}5 - 1{,}2 : 0{,}3$ $\quad \frac{275}{100} + \frac{48}{100} \cdot \frac{25}{10} - \frac{12}{10} : \frac{3}{10}$

$\frac{275}{100} + \frac{1200}{1000} - 4 = \frac{275 + 120}{100} - 4 = \frac{395}{100} - \frac{400}{100} = \frac{-5}{100} = -0{,}05$

25) $\frac{\frac{2}{100}+\frac{3}{100}+\frac{4}{100}}{\frac{4}{1000}+\frac{5}{1000}+\frac{8}{100}} = \frac{2}{100}\cdot\frac{1000}{42} + \frac{3}{100}\cdot\frac{1000}{5} + \frac{4}{100}\cdot\frac{100}{82}$

$5+6+\frac{1}{2} = 11+\frac{1}{2} = \frac{23}{2} = 11{,}5$

26) 97 asaldır

27)
360	2
180	2
90	2
45	3
15	3
5	5
1	

$2^3 \cdot 3^2 \cdot 5$

28) $756 = 2^2 \cdot 3^3 \cdot 7$ $2+3+7=12$

29) $360 = 2^3 \cdot 3^2 \cdot 5$ Pozitif bölen $\rightarrow (3+1)(2+1)(1+1) = 24$

30) $N = 2^a \cdot 3^2 \cdot 5$ $(a+1)\,3 \cdot 2 = 24$ $a = 3$

31) $72 = 2^3 \cdot 3^2$ $2 \rightarrow 1,2,3$ $3 \rightarrow 0,1,2$ $3 \cdot 3 = 9$

32) $144 = 2^4 \cdot 3^2$ $2 \rightarrow 0,2,4$ $3 \rightarrow 0,2$ $3 \cdot 2 = 6$

33) EBOB(84,126) $84 = 2^2 \cdot 3 \cdot 7$, $126 = 2 \cdot 3^2 \cdot 7$ $2 \cdot 3 \cdot 7 = 42$

34) EBOB(72,120,168) $72 = 2^3 \cdot 3^2$, $120 = 2^3 \cdot 3 \cdot 5$, $168 = 2^3 \cdot 3 \cdot 7$ $2^3 \cdot 3 = 24$

35) EKOK(84,126) $84 = 2^2 \cdot 3 \cdot 7$, $126 = 2 \cdot 3^2 \cdot 7$ $2^2 \cdot 3^2 \cdot 7 = 252$

36) EKOK(18,24,30) $18 = 2 \cdot 3^2$, $24 = 2^3 \cdot 3$, $30 = 2 \cdot 3 \cdot 5$ $2^3 \cdot 3^2 \cdot 5 = 360$

37) EKOK(12,18) = 36 36, 72, 108

38) EBOB(48,b) = 12 EKOK(48,b) = 240 $48 \cdot b = 12 \cdot 240$ $b = 60$

39) EBOB(a,b) = 6 EKOK(a,b) = 180

40) EBOB(18,30) = 6 EBOB(54,30), EKOK(

$54 = 2 \cdot 18$ $30 = 3 \cdot 10$ EBOB(54,

41) EBOB(84,126) = 42

42) EBOB(48,60) = 12

43) EBOB(60,84) = 12

44) EBOB(96,120) = 24

45) EKOK(12,18,30)

46) EKOK(8,12) = 24

47) EKOK(6,8) = 24

48) EKOK(4,6,9) =

49) EBOB(252,360)

50) EKOK(18,24,12)

51) $540 = 2^2 \cdot 3^3 \cdot 5$

52) EBOB(A,B) = 2^2 ·
EKOK(A,B) = 2^3
360/12 = 30

$\frac{10}{\frac{1000}{42}} + \frac{3}{140} \cdot \frac{\frac{280}{5}}{} + \frac{4}{140} \cdot \frac{140}{202}$

7) 360 | 2
180 | 2
90 | 2
45 | 3
15 | 3
5 | 5
1 | $2^3 \cdot 3^2 \cdot 5$

3+1)(2+1)(1+1) = 24

a = 3

3.3 = 9

3.2 = 6

3.7 = 42

$= 2^3 \cdot 3 \cdot 7$

$2^3 \cdot 3 = 24$

$3^2 \cdot 7 = 252$

2.3.5

$= 360$

39) EBOB(a,b) = 6 $\quad a = 6x \quad b = 6y$ $\quad 36xy = 6.100 \quad xy = 30 \quad 5 \cdot 6 \quad a = 30 \quad b = 36$
EKOK(a,b) = 180

40) EBOB(18, 30) = 6
EBOB(54, 90), EKOK(54, 90) = ?
$54 = 3.18$ $\quad$ EBOB(54, 90) = 3.6 = 18
$90 = 3.30$ $\quad$ EKOK(54, 90) = 3.90 = 270

41) EBOB(84, 126) = 42 $\quad 84/42 = 2 \quad 126/42 = 3 \quad 3+2 = 5$

42) EBOB(48, 60) = 12 $\quad 48/12 = 4 \quad 60/12 = 5 \quad 4+5 = 9$

43) EBOB(60, 84) = 12 $\quad 2.60 + 2.84 = 288 \quad 288/12 = 24$

44) EBOB(96, 120) = 24 $\quad 96/24 = 4 \quad 120/24 = 5 \quad 4.5 = 20$

45) EKOK(12, 18, 30) = $2^2 \cdot 3^2 \cdot 5 = 180 \quad 9+8 = 12$

46) EKOK(8, 12) = 24 $\quad 24a + 5 = k \quad a = 4 \quad k = 101$

47) EKOK(6, 8) = 24 $\quad \frac{24^2}{6.8} = \frac{576}{48} = 12$

48) EKOK(4, 6, 9) = 36 $\quad \frac{36^3}{4.9.6} = \frac{\overset{6}{\cancel{36}} \cdot \overset{9}{\cancel{36}} \cdot 36}{4.6.9} = 36.9 = 216$

49) EBOB(252, 360) = $2^2 \cdot 3^2 = 36 \quad 252/36 = 7 \quad 360/36 = 10 \quad \frac{7}{10}$

50) EKOK(18, 24, 12) = 72 $\quad \frac{20}{72} + \frac{21}{72} - \frac{6}{72} = \frac{35}{72}$

51) $540 = 2^2 \cdot 3^3 \cdot 5 \quad k = 3.5 = 15$

52) EBOB(A,B) = $2^2 \cdot 3 = 12$
EKOK(A,B) = $2^3 \cdot 3^2 \cdot 5 = 360$
$360/12 = 30$

53) EBOB(36, 54) = k = 9
EKOK = $\frac{(24.40)}{9} = 120$

ALIŞTIRMALAR

1) $5-2\cdot\underbrace{[3-(2-6)\cdot 2]}_{-22}+\underbrace{18:(-3)}_{-6}=-23$

2) $2x-\{3-\underbrace{[x-2(x+4)]}_{x-2x-8}\}=2x-(3+x+8)=2x-3-x-8=x-11$

3) $\frac{49-5}{4}+1=12 \qquad \frac{49+5}{2}\cdot 12^6=324$

4) $(231)_4 \qquad 16\cdot 2+12+1=32+13=45$

5) $\left[\left(\frac{3}{4}-\frac{1}{6}\right):\frac{7}{12}\right]+\frac{3}{2}=\frac{7}{12}\cdot\frac{12}{7}+\frac{3}{2}=1+\frac{3}{2}=\frac{5}{2}$

6) $\frac{15}{10}:\frac{12}{100}-\frac{25}{100}\cdot 8=\frac{15^3}{10}\cdot\frac{100^{10}}{12_2}-\frac{200}{100}=15-2=13$

7) $420=2^2\cdot 3\cdot 5\cdot 7$ 8) $180=2^2\cdot 3^2\cdot 5 \qquad 3\cdot 32=18$

9) $(a+1)\cdot 2\cdot 2=20 \qquad a=4$ 11) $60\,b=15.120 \qquad b=45$

10) $96=2^5\cdot 3 \qquad EBOB=48$
$144=2^4\cdot 3^2 \qquad EKOK=288$

12) $EBOB(72,120)=24 \qquad \frac{72}{24}=3 \qquad \frac{120}{24}=5 \qquad 5+3=8$

13) $EBOB(48,72)=24 \qquad 2.48+2.72=240 \qquad 240/24=10$

14) $EKOK(15,20)=60 \qquad 10.10+01.00=11.10$

15) $EKOK(3,4,6)=12 \qquad \frac{12^3}{4\cdot 3\cdot 6}=\frac{12.12.12^2}{4\cdot 3\cdot 6}=24$

16) $6x \qquad xy=12$
$6y \qquad EBOB(x,y)=1 \qquad 6(3+4)=42$

Globally Yours